



Dr. N.G.P. INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

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Recognized by UGC & Accredited by NAAC with A+ and NBA (BME, CSE, ECE, EEE, IT and Mech)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Rec. Name :
TLP

22UAD915 & Ethics in Data Science

ASSIGNMENT 1

S.No	Questions	CO/ Knowledge level	Marks
1.	Two employees at a company disagree on whether an algorithm's outcome is "right," even though it complies with all regulations. Analyze the ethics-vs-law conflict illustrated in this situation.	CO1 / K4	8
2.	A company introduces a mandatory case-study training program for new data science hires before allowing them to handle live projects. Explain the purpose and benefits of this practice.	CO1/ K3	8
3.	A data science consultant is hired to build a predictive policing tool. The client wants historical arrest data used directly, even though the consultant knows it reflects past biased policing practices, which the IEEE/ACM codes caution against perpetuating. a) Identify the relevant clauses from a professional code of ethics (e.g., ACM Code) that apply to this situation. (3 marks) b) Discuss how formal codes guide professionals in refusing or renegotiating unethical client requests. (2 marks) c) Evaluate what a professional should do if the client insists on proceeding despite these concerns. (3 marks)	CO2/ K3	8
4.	A smart city project aggregates data from traffic cameras, public Wi-Fi logs, and transit cards to optimize urban planning. Researchers later show that combining these "anonymized" datasets can re-identify specific individuals' daily routines. a) Explain the concept of the "mosaic effect" in big data privacy and how it applies here. (3 marks) b) Discuss why traditional anonymization techniques often fail in big data contexts. (2 marks) c) Recommend privacy-by-design measures the city should implement going forward. (3 marks)	CO2/ K3	8

5.	A healthcare startup develops a mobile application that monitors users' heart rate, sleep patterns, and daily activities using wearable devices. The company informs users that the data will improve health recommendations. Later, the company sells anonymized health data to pharmaceutical companies without explicitly informing users. a) Identify the privacy issues involved in the scenario. b) Explain whether user consent was ethically sufficient. c) Discuss the ethical responsibilities of the company. d) Suggest two measures that could improve user privacy.	CO3/ K3	8
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Dr.N.G.P. Institute of Technology, Coimbatore			
Department: Artificial Intelligence and Data Science			
Course code and Course Name		22UAD915 & Ethics in Data Science	
Sem/Year		VII / IV & V / III	
Programme		B.Tech – AI &DS	
Batch		2023-2027 & 2024-2028	
Faculty Name			
Date of issue:	03.08.2026	Submission Due:	12.08.2026

Instructions to students

- ANSWER NEATLY AND LEGIBLY on A4 sheets only and not in sheets torn from a book.
- Sketch diagrams wherever relevant. Explain your notations explicitly and clearly.
- An incomplete assignment is not acceptable for submission.
- Once you submit your assignment, you will be expected to answer all the questions there INDEPENDENTLY.
- You may be asked to answer any question about the assignment in the class.
- Leave a clear left margin on every page.
- On the top of the first page, write your Name, Class, Section, Roll No., Due Date and Date of Submission.
- Staple (do NOT pin or clip) the assignment solution sheets cleanly at the left-hand top corner.